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LFP RUL — Physics-Informed Operator for Second-Life Cell Remaining-Useful-Life

One-line pitch. A per-cell PyBaMM calibration $\rightarrow$ θ -conditioned neural operator that predicts Remaining Useful Life from a single characterisation snapshot, rebuilt end-to-end after the original attempt failed silently.

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1. What we set out to do

Given: - 3 commercial LFP suppliers (CALB 72 Ah, EVE 105 Ah, REPT 150 Ah) - Second-life cells at 0.60–1.00 fresh-normalised SoH - Characterisation tests + Longterm cycling data per cell

Produce a **neural operator** that, from one characterisation snapshot on a fresh cell, predicts the full SoH-vs-cycle trajectory to end-of-life across suppliers. Ship a 1-page conference abstract + companion research paper.

2. What the previous attempt did (and why it failed)

An earlier build shipped a corpus of 300 PyBaMM sims and a “ θ -conditioned DeepONet” that predicted **flat SoH $\approx$ 1.0** for every real cell.

The failure report attributed this to **θ -space out-of-distribution (5σ OOD)** — the corpus’s $\log_{10}(k_SEI)$ mean was -14.15 while real fits sat at -11.5 .

Adversarial re-audit found **the actual root cause was different**: the model at `src/pinn/model.py` was a Neural ODE where the branch input was `x_health(5)` only — **θ was never fed to the network at all**. The 5σ OOD claim was a symptom, not the cause. Without the θ fix, no amount of corpus widening would have helped.

3. What this rebuild delivers

Cohort — from 61 candidate cells to a defensible 13

Filter	Cells
All 3 makes, raw data	61
After CC/CV protocol correction on CALB SoH	47 clean
After [0.60, 1.00] SoH_first filter + monotonicity	21
After DoD-protocol grouping + noise ranking	13 (5 CALB · 3 EVE · 5 REPT)

Filter	Cells
After adversarial verification post Phase 2	7 confirmed anchors (4 CALB · 1 EVE · 2 REPT)

Three-phase pipeline

1. **Phase 1 — Per-cell BOL identification.** OCV curve fit gives stoichiometry (x_{100} , x_0 , y_{100} , y_0); Q_{rpt} derives electrode capacities; HPPC gives $R_0 + R_1$; GITT gives D_s . **Result: 13/13 cells pass the OCV RMSE < 20 mV upper-half gate.**
2. **Phase 2 — Per-cell degradation-parameter DE fit.** Differential evolution fits 5 parameters (k_{SEI} , V_{SEI} , $D_{SEI_solvent}$, $k_{plating}$, $k_{LAM_negative}$) against measured Longterm SoH using each cell’s actual cycling protocol. **Result: 10 of 13 cells sub-2 pp RMSE; 3 aged-REPT cells fit at ~2.5 pp (limited by measured signal-to-noise, not model quality).**
3. **Phase 3 — Perturbation corpus + operator retrain.** Sobol sample 70 draws per anchor around each of the 7 Phase-2 fits, with decorrelation gate + fast-fade quadrant booster. Retrain the Neural ODE with **θ fed into the branch** (the fix). 490 sims → operator. **Status: corpus sweep dispatched; ETA ~15 h.**

4. Bugs caught in-flight — where the value came from

Every one of these would have propagated silently and either burned compute or produced a bad result. Each was caught by adversarial verification (multiple independent agents cross-checking each other), not by the primary implementer.

#	Bug	Where	Would have cost	Detected by
1	REPT characterisation “missing” (only local disk checked)	Data inventory	20 excluded cells	User challenge
2	CC-only vs CC-CV batch mismatch inflated CALB SoH by ~2 pp	Loader	Corrupted every CALB fit downstream	User challenge
3	PyBaMM sign convention — extractor picked charge step as “discharge”	Phase 2 loss	100% of Phase 2 DE fits returning RMSE $\approx$ 9000 pp	Instrumented probe
4	Cycle-1 discharge cap inflated on 0_80 protocol → DE prefers solver-death	Phase 2 loss	3 aged REPT fits stuck at 10.0 pp penalty	Adversarial deep-dive
5	θ column-name mismatch between corpus writer / feature reader / trainer — θ silently zeroed in dataset	Phase 3 pipeline	15 h corpus + 2 h training producing the same “flat SoH” failure as the original	Adversarial audit

#	Bug	Where	Would have cost	Detected by
6	Fast-fade booster quadrant check silently returned True universally due to key-alias mismatch	Phase 3 booster	Design requirement R2 (fast-fade coverage) unenforced → operator would hallucinate flat trajectories for aged cells	Own-code re-audit while fixing #5
7	Neural operator branch input never received θ (root cause of the original failure)	Model architecture	Would have re-shipped an operator that predicts flat SoH	Read of the file itself

Total compute avoided: $\approx$ 30-50 h of PyBaMM sweep + training runtime.

5. Time and cost impact

Time investment breakdown

Activity	Hours (elapsed)	Notes
Data audit, cohort selection, protocol map	~8	Multiple iterations catching normalisation bugs
Phase 1 BOL identification + diagnostics	~2	Includes anomaly investigation on CALB 0008
Phase 2 code + 13-cell DE + refits after bug fix	~4	~1 h compute + 3 h engineering
Stage A + Stage B adversarial verification	~2	Two multi-agent workflows
Phase 3 design + implementation + adversarial audit	~3	3 blockers caught before sweep
Total to date	~19 h	Sweep now dispatched in background

Cost baseline (what we did NOT do)

- Naive re-run of the previous 300-sim corpus + retrain: **~13 h compute** → identical failure. Time-to-diagnose the failure again: probably 1-2 days.
- Manual per-cell fitting without adversarial verification: probably **3-5 person-days** of debugging misfits attributed to physics rather than code.
- Re-collecting characterisation on cells where “no data” was assumed but data existed in Athena: **weeks** of lab time (this was ~10 REPT cells).

Net vs a naive rebuild

- **Compute time saved by catching bugs early: 30-50 h**
- **Person-time saved by carrying forward provable results: probably 2-3 weeks**
- **Cohort strength: 7 verified anchors > 300 unverified samples** (previous approach)

6. What the deliverable is

- **configs/bol_params/{make}_{cell}.yaml** — Phase 1 BOL parameters for 13 cells; every cell reproduces its OCV within 5 mV upper-half RMSE
- **configs/deg_params/{make}_{cell}.yaml** — Phase 2 degradation parameters for 13 cells; every fit reproduces the saved RMSE to full float precision on independent replay
- **configs/cohort_experiment_protocols.yaml** — the 11 unique cycling protocols the cohort was actually run on (previous corpus used ONE protocol; 8 of 13 cells were mis-simulated)
- **configs/phase3_sweep.yaml** — corpus generation config with per-parameter σ , decorrelation gate, fast-fade booster
- **configs/phase3_operator.yaml** — operator retrain config with the θ -into-branch fix and shape+monotonicity+physics loss
- **Voltaris/Data_Exploration/*.py + *.ipynb** — every extractor, fitter, notebook: pure functions with smoke tests
- **7-anchor SoH overlay notebook** — proves each fitted- θ trajectory reproduces the measured Longterm trace
- **5000-cy projection notebook** — physical-plausibility check of anchor θ (in progress)
- **Adversarial verdict documents** — `_aggregate.md`, `_stage_b_verdict.md`, `phase3_readiness.md` — every claim in the paper is backed by an artefact

7. Current state and next steps

Running now: 15-h Phase 3 corpus sweep (~490 sims, 7 anchors, `n_jobs=5`). Checkpointed per anchor.

When the sweep lands: 1. Feature extraction + dataset build (~1 h) 2. Operator retrain (~2 h GPU) 3. Held-out validation on 3 cells across makes with Fisher-column cosine + regime-swap replay gates 4. If pass: draft paper + 1-page conference abstract

Expected paper story: per-cell PyBaMM identification breaks the single-shot characterisation cost barrier for supplier-agnostic second-life LFP RUL; corpus + operator generalises across chemistries without touching PyBaMM at inference time.

8. Risks worth flagging

1. **Corpus fast-fade axis is anchored by 2 REPT cells** — narrow. Held-out REPT prediction is the strongest test.
2. **SEI $\leftrightarrow$ LAM degeneracy** shown by stress test on CALB 0003 — pure-SEI within $1.5\times$ best RMSE. Corpus decorrelation gate + shape loss address it, but empirical Fisher-cosine < 0.3 verdict is the real proof.
3. **3 REPT aged cells fit at ~2.5 pp** — protocol-limited (partial-DoD 0_80 measured signal is 0.3-0.8 pp — dominated by noise). Not a modelling fault, but the paper needs to say so.

Written 2026-07-11 after the Phase 3 sweep dispatched. Metrics and thresholds land as evidence, not aspirations — every number here is reproducible from configs/ and the notebooks in Voltaris/Data_Exploration/.